

environment, but typically include sufficient CPU power, adequate RAM, and ample storage space to accommodate backup data and the Bacula catalogue database. A stable network connection is essential for seamless communication between Bacula components.

On the software side, a supported database system like MySQL, PostgreSQL, or SQLite is required for catalogue management, along with the necessary dependency libraries such as OpenSSL for encryption and zlib for compression. Backup media options include disks, tapes, or cloud storage, depending on the specific needs of your organisation.

For the setup we will be using any RPM-based Linux distribution with MySQL installed on it.

To install the Bacula software suite, an open source enterprise-level backup solution, use the command given below:

```
[root] dnf -y install bacula-director bacula-storage bacula-console bacula-client
```

Preparing and setting up MySQL for Bacula

To set Bacula to use the MySQL library, you can run the command:

```
[root] alternatives --config libbaccats.so
```

Let's create the Bacula database user and tables with the following scripts, or you can do it manually:

```
/usr/libexec/bacula/grant_mysql_privileges
/usr/libexec/bacula/create_mysql_database -u root
/usr/libexec/bacula/make_mysql_tables -u OSFY
```

Use the following command to go to the *mysql* prompt and then create a database for OSFY with the permissions mentioned below:

```
[root]mysql
[mysql]create database OSFY
[mysql] grant all privileges on OSFY.* to OSFY@'127.0.0.1'
identified by 'OSFY';
[mysql] grant all privileges on OSFY.* to OSFY@'%' identified
by 'OSFY';
[mysql] flush privileges;
```

Next, you can create the table for the OSFY database:

```
[root]/usr/libexec/OSFY/make_mysql_tables
```

Now we need to configure Bacula. To set the Bacula directory, use the following command:

```
[root] vi /etc/bacula/bacula-dir.conf
# line 25 : set password for Director
Password = "OSFY"
# line 110 : add (enable RAR or ZIP compress)
Compression = RAR
# line 126 : specify backup target directory
File = /home
# line 178 : password for File daemon
Password = "OSFY"
# line 204 : change to FQDN of your host according to the
note
Address = opensourceforu.com

# line 206 : password for Storage daemon
Password = "OSFY"
# line 243 : password of Mysql bacula user
dbname = "OSFY"; dbuser = "OSFY"; dbpassword = "OSFY"

# line 304 : volume retention term
Volume Retention = 300 days
# max volume size
Maximum Volume Bytes = 75GB
# max volume number
Maximum Volumes = 150
# prefix for volume files
Label Format = "OSFY-"

# line 322 : tray-monitor password
Password = "OSFY"
```

The following command is for the Bacula console:

```
[root] vi /etc/bacula/bconsole.conf
# password for Director
Password = "OSFY"
[root] systemctl enable --now bacula-dir
```

Configuring Bacula storage

Use the following command to configure Bacula storage:

```
[root]vi /etc/bacula/bacula-sd.conf
```

The file */etc/bacula/bacula-sd.conf* is the main configuration file for the storage daemon. The two specified changes in the file are:

```
# line 31 : set password for Storage daemon
Password = "OSFY"
# line 40 : tray-monitor password
Password = "OSFY"

[root] systemctl enable --now bacula-sd
```